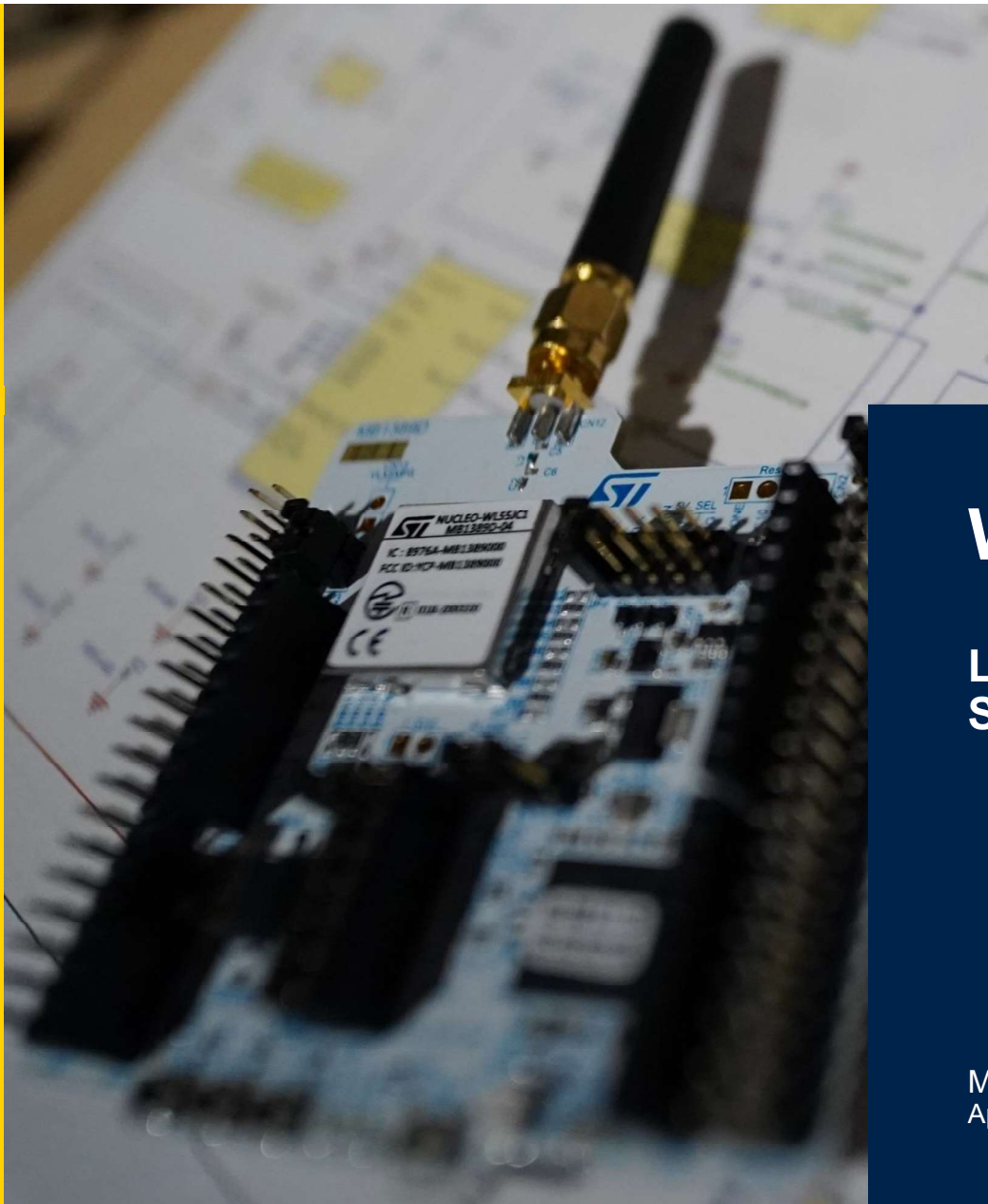




life.augmented

Workshop LAB#1

LED Blinky via
STM32Cube ecosystem

Minku Yeo
Applications Engineer



Glossaries

- LoRa uses license-free sub-gigahertz radio frequency band
 - High Frequency band = 868MHz to 915MHz
 - Low Frequency band = 169MHz and 433MHz, 470MHz
 - North America - 915 MHz band (High Frequency band)
- NUCLEO-WL55JC1 - HF High Frequency bands
VS
NUCLEO-WL55JC2 - LF Low Frequency bands



Requirements

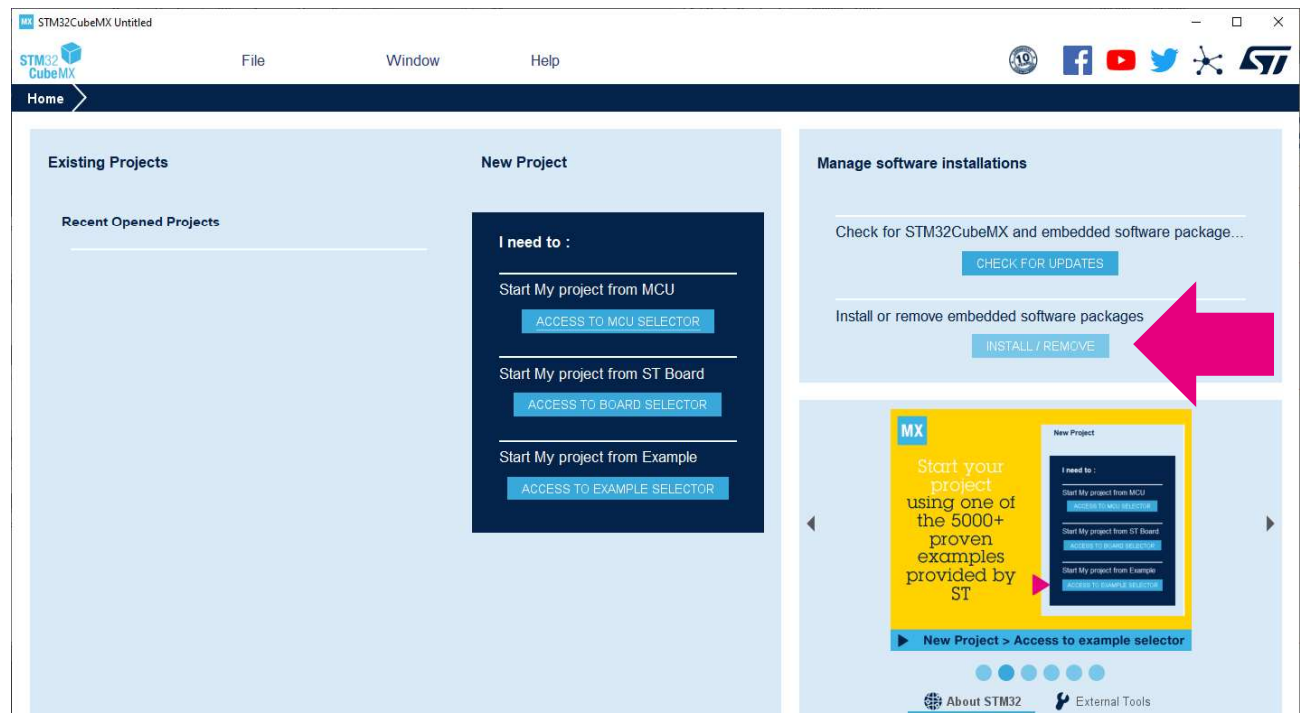
- Hardware:
 - One NUCLEO-WL55JC1 HF boards
 - Micro USB (A-Male to Micro B) cable
- Software:
 - STM32CubeMX v6.3.0 or above
 - STM32CubeIDE v1.7.0 or above



USB

Download the STM32CubeWL package

- Download the firmware package via STM32CubeMX v6.3.0
- Press Install/Remove button



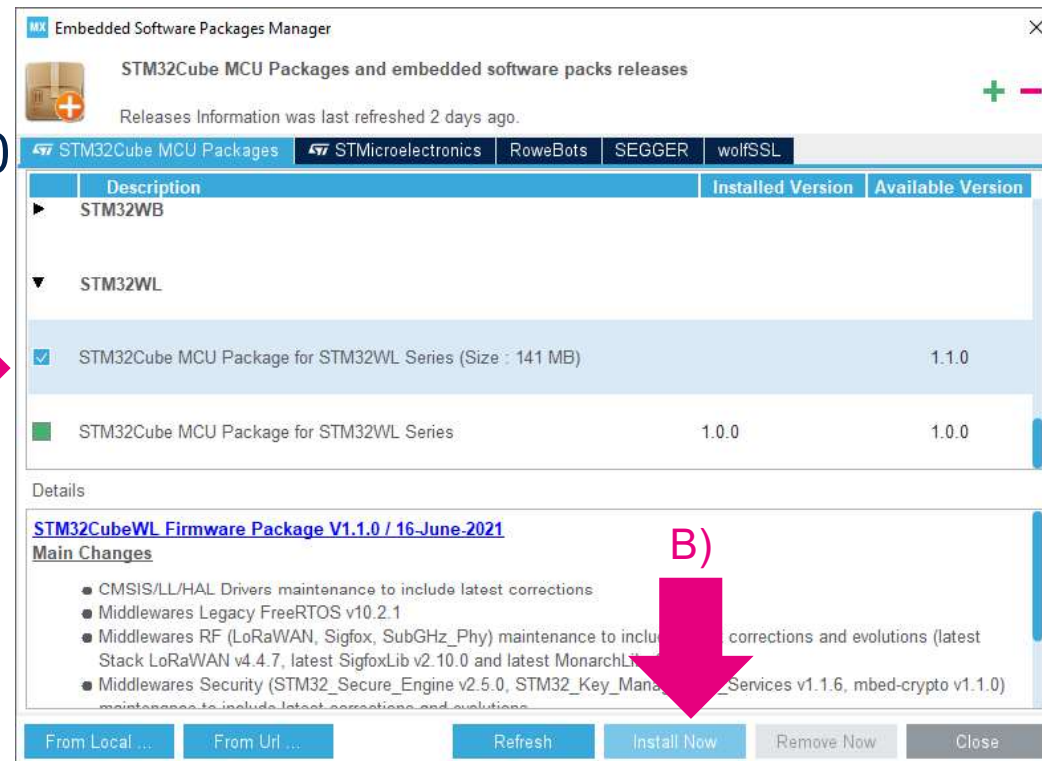
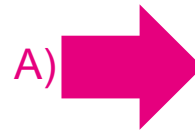
Download the STM32CubeWL package

- Scroll to the STM32Cube MCU package - STM32WL

- A) Check the checkbox of the latest v1.1.0

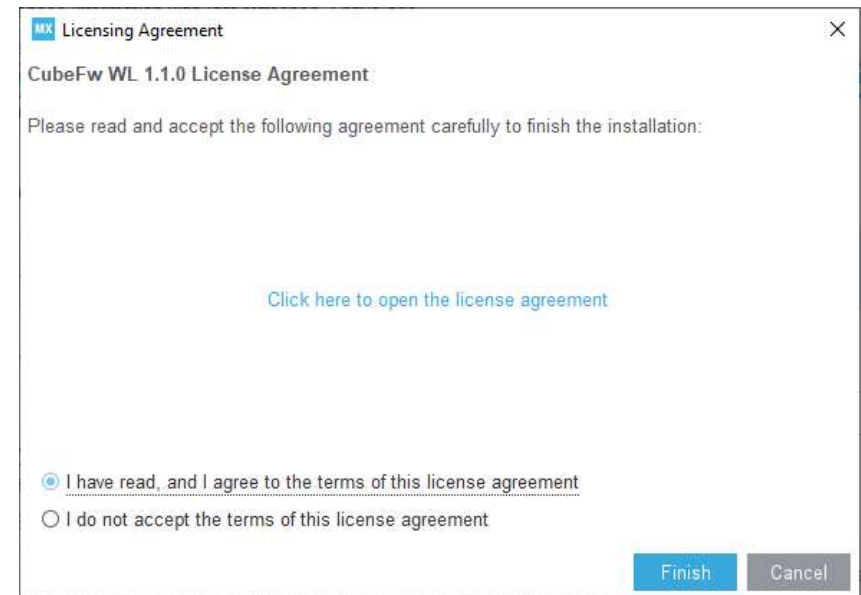
- B) Click install now

- Turn off network firewall for proper installation



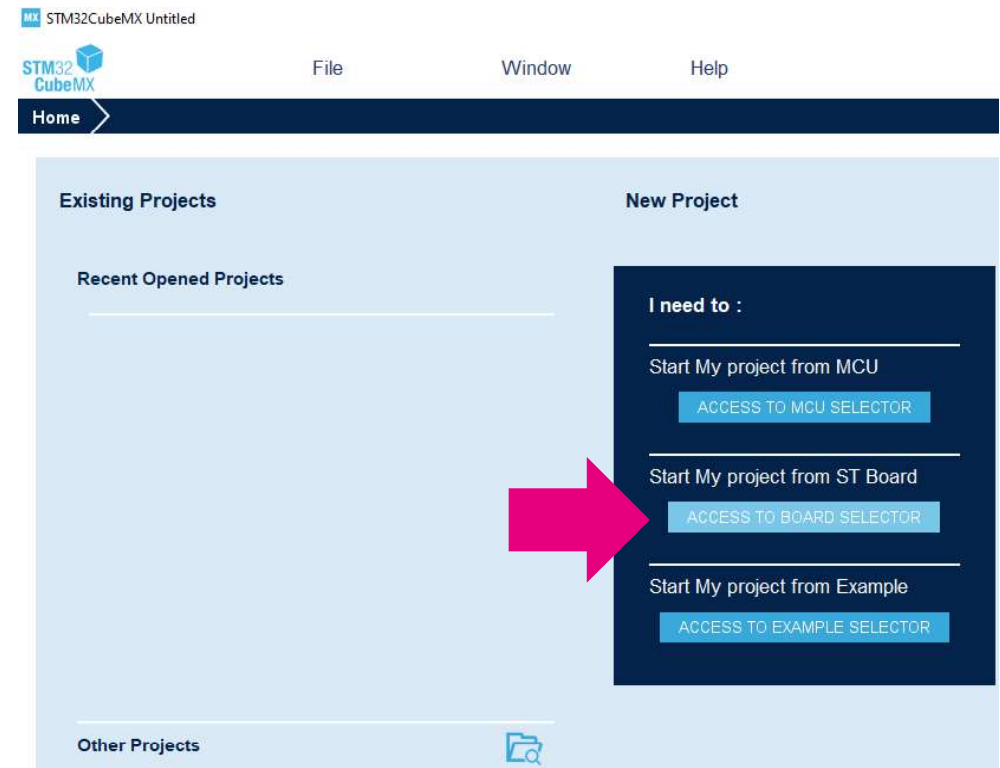
Download the STM32CubeWL package

- Accept the agreement to start the installation
- **PLEASE use the latest STM32CubeMX v6.3.0+**
If you are using an old version, like v6.2.0,
you might see different pop-ups



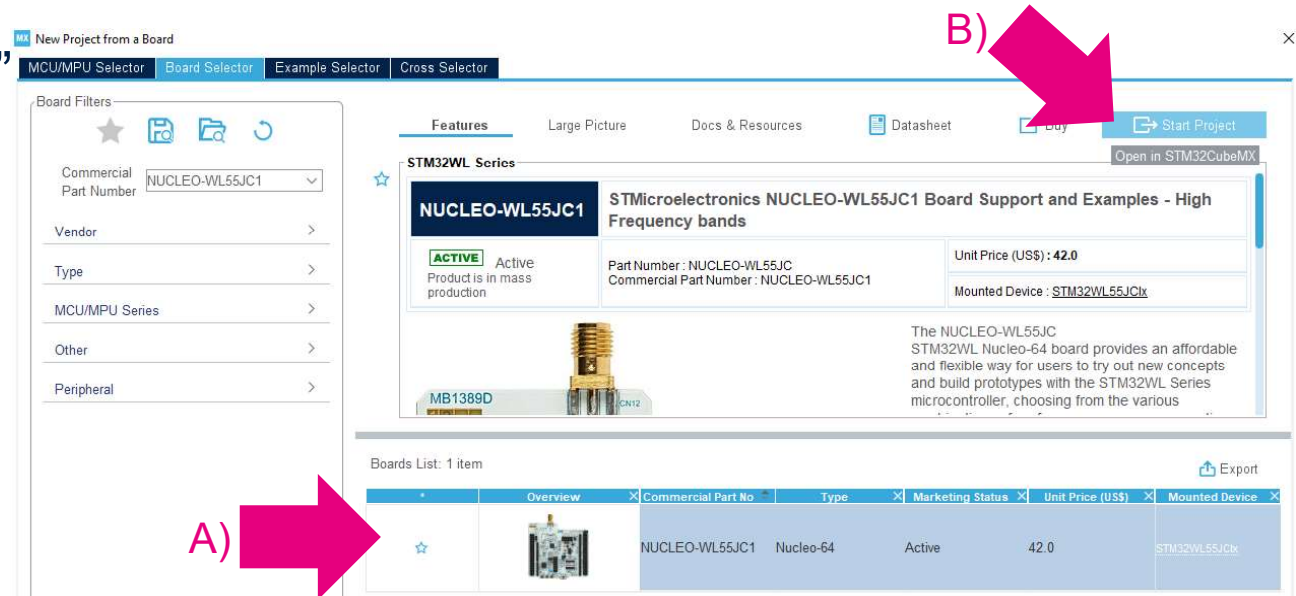
Create a project via STM32CubeMX

- Click “Access to board selector”
- Network connection is required when using this menu



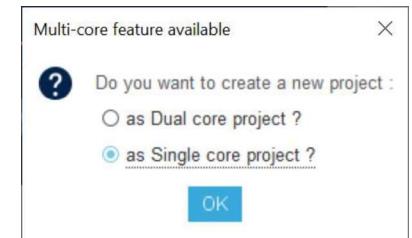
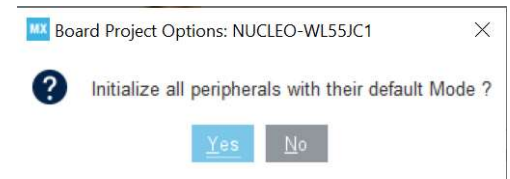
Create a project via STM32CubeMX

- Click “Access to board selector”
- Type NUCLEO-WL55JC1
- A) Select and
B) click "Start Project"



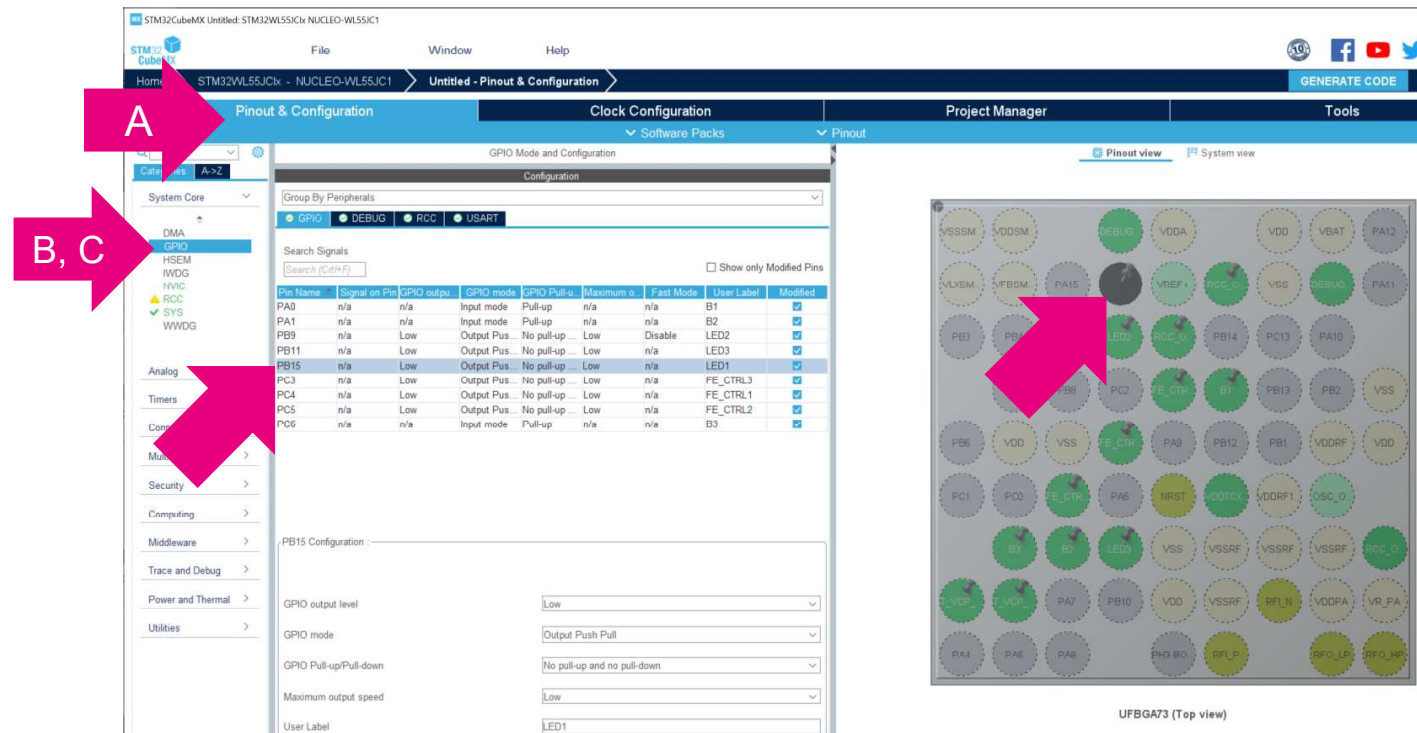
Create a project via STM32CubeMX

- Click “Yes” for initializing the default peripherals
- Select “Single core project” and “OK”
- NOTE: if you use STM32CubeMX v6.2.1 or older, these dialog boxes will look different.
This lab used STM32CubeMX v6.3.0



Create a project via STM32CubeMX

- View the LED pins at the (A) Pinout & Config – (B) System core – (C) GPIO
- Highlight a pin, like PB15/PB11/PB9
You'll notice a pin toggling from the Pinout view



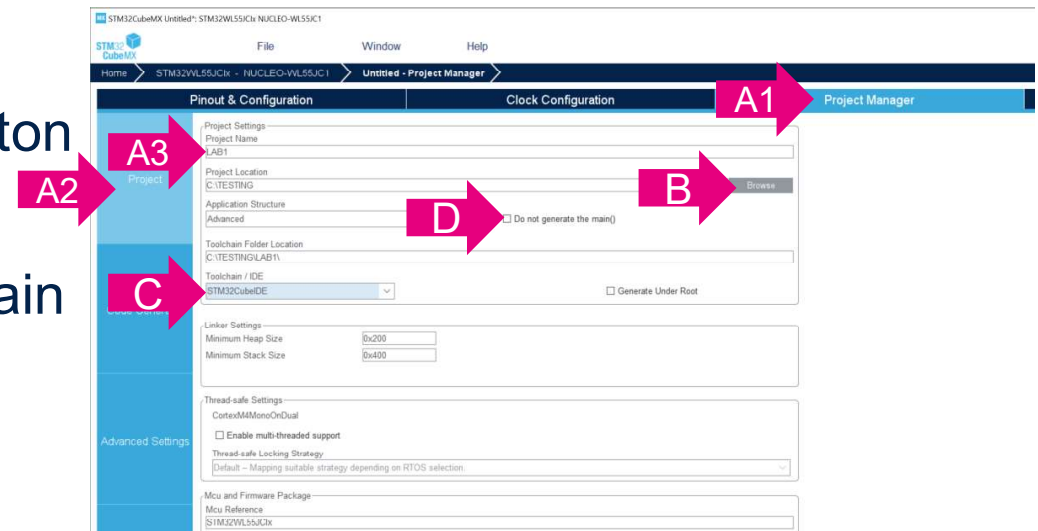
Create a project via STM32CubeMX

A. From the Project Manager - Project tab, name the project, like LAB1

B. Select the directory, use the Browse button

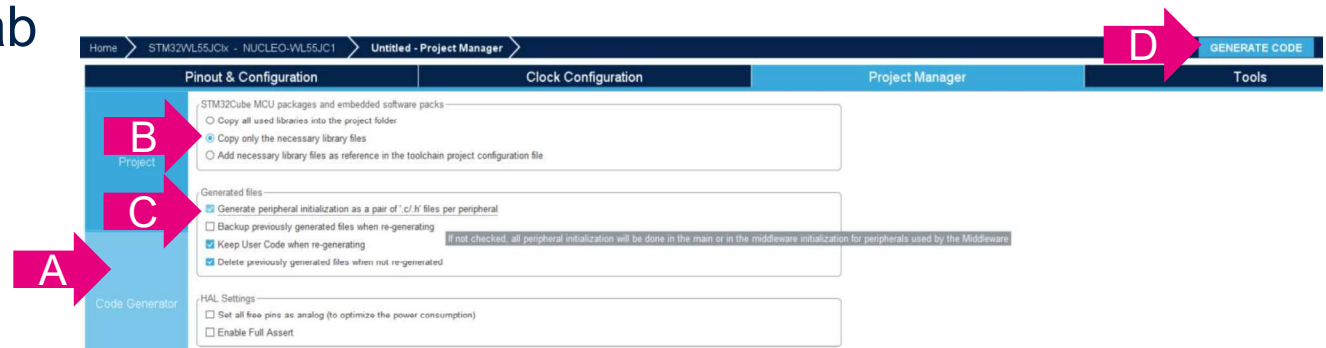
C. Choose STM32CubeIDE as your toolchain

D. Uncheck "Generate Under Root"



Create a project via STM32CubeMX

A. Select the Code generator tab



B. Select “Copy only the necessary library files”

C. Check “Generate peripheral initialization as a pair”

D. Click “Generate Code”

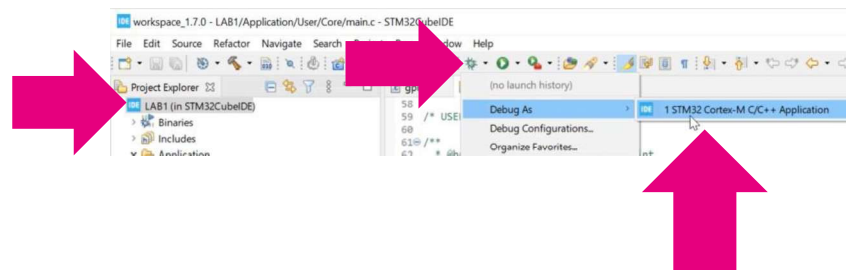
Create a project via STM32CubeMX

- Wait for the progress bar
- Click "Open Project"



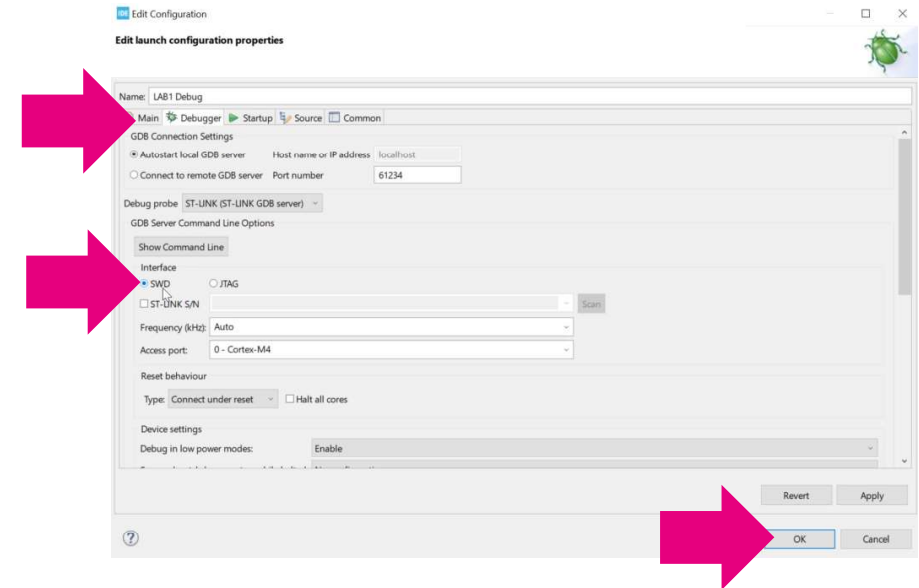
STM32CubeIDE - debug

- Connect the NUCLEO-WL55JC1 to your PC first.
- Then click the dropdown arrow from the bug icon - debug as - STM32 application
- Your project (LAB1) has to be still highlighted while doing this



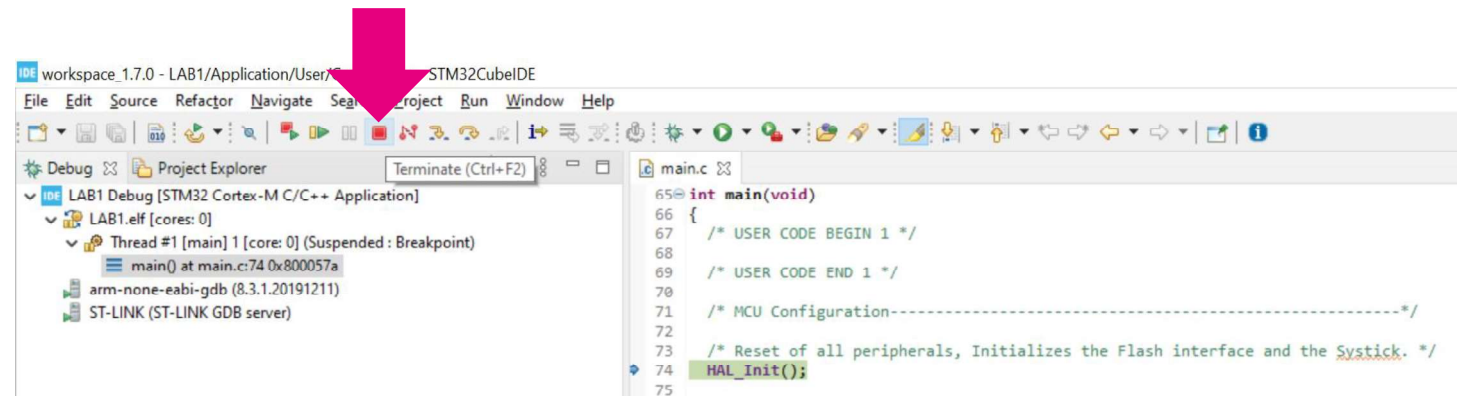
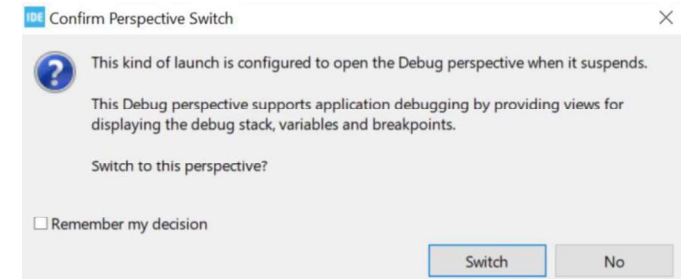
STM32CubeIDE - debug

- Click the "Debugger" tab
- From "Interface" make sure SWD is selected.
- Click OK



STM32CubeIDE - debug

- Click “Switch” and use the debugging feature
- Click the "Terminate" button to start editing the code



STM32CubeIDE - coding

- Add your code, for instance, toggle LEDs in the main function's while loop
- Save, build, and debug to see the blinking LEDs

```
while (1)
{
    /* USER CODE END WHILE */
    HAL_GPIO_TogglePin(LED1_GPIO_Port, LED1_Pin);
    HAL_Delay(500);
    HAL_GPIO_TogglePin(LED2_GPIO_Port, LED2_Pin);
    HAL_Delay(500);
    HAL_GPIO_TogglePin(LED3_GPIO_Port, LED3_Pin);
    HAL_Delay(500);
    /* USER CODE BEGIN 3 */
}
```

